

Mounir Arayssi

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Education

North Carolina State University

B.S. Computer Science

Raleigh, NC

Graduated May 2026

Experience

Dar Al Handasah

IT Infrastructure Intern

Beirut, Lebanon

June 2025 – July 2025

- Improved infrastructure scalability and reduced costs by 40% by consolidating servers and deploying a VMware ESXi environment supporting 15+ data-intensive applications
- Enabled non-technical users to query structured data independently with 80% accuracy by building a Python NLP-to-SQL pipeline integrating REST APIs with a MySQL database

Paragon Shift

Data Engineering Intern

Cleveland, OH

May 2024 – July 2024

- Gathered requirements from non-technical users and built a config-driven Python CLI using pandas to reconcile CSV exports through normalization, deduplication, composite-key matching, and row-level comparisons
- Automated data quality controls identifying mismatched values, exact duplicates, duplicate IDs, and system-exclusive records, producing structured exception reports for QA, auditing, and stakeholder review
- Developed Power BI dashboards for curated datasets, improving visibility into data quality trends, operational KPIs, and recurring reporting needs for business stakeholders

NC State University College of Natural Resources

IT Intern

Raleigh, NC

Aug. 2023 – May 2024

- Supported high-availability Windows and Linux systems serving 2,000+ users, troubleshooting incidents and maintaining reliable access to critical campus infrastructure

Projects

Lab ETA Prediction System | *Python, LightGBM, CatBoost, FastAPI, React, Docker*

- Partnered with NetApp engineers to translate lab scheduling and capacity-planning needs into an ML pipeline processing 90,000+ job records, achieving a 79% MAE improvement over the baseline
- Engineered queue-state and temporal features and implemented conformal calibration with chronological train-test splits, preventing data leakage and delivering uncertainty-aware prediction intervals
- Deployed a containerized full-stack application using React, FastAPI, and Nginx with CI/CD automation, providing a REST API and interactive interface for real-time job completion predictions

NBA Prediction Analytics Platform | *Python, scikit-learn, PostgreSQL, SQLAlchemy, Streamlit, Pandas*

- Achieved 70% winner accuracy and 9.4-point MAE across 3,700+ NBA games using chronological train-test splits, outperforming the home-team baseline by 14 percentage points
- Engineered a gradient-boosting ensemble with 50+ features spanning Vegas spreads, injury load redistribution, playoff efficiency, team form, rest, and matchup trends
- Automated a daily pipeline ingesting NBA schedules, box scores, ESPN injury reports, and DraftKings odds; retraining models; storing predictions in Neon PostgreSQL; and serving results through Streamlit

Flight Intelligence Platform | *Python, Apache Airflow, BigQuery, GCS, Docker, PostgreSQL*

- Built an hourly ELT pipeline ingesting 1,800+ live U.S. flight-position records per run from a public ADS-B API, staging raw snapshots in GCS and appending them to a BigQuery historical dataset
- Containerized a distributed six-service Airflow stack using Docker Compose, CeleryExecutor, Redis, and PostgreSQL with health checks and dependency ordering for reproducible deployment

Leadership

Lebanese Club at NC State

President

Raleigh, NC

Aug. 2023 – May 2026

- Led planning and stakeholder coordination for 15+ cultural and professional events serving 35+ active members, establishing partnerships with campus organizations and local communities to expand programming reach

Technical Skills

Languages: Python, SQL, Java, JavaScript

Analytics & Business Intelligence: Power BI, pandas, NumPy, KPI Reporting, Dashboard Development, Data Reconciliation, Data Quality & Validation

Data Engineering & Cloud: Apache Airflow, ETL/ELT Pipelines, Workflow Orchestration, GCP, GCS, BigQuery, Redis

Databases: PostgreSQL, MySQL, Neon, SQLAlchemy

Machine Learning: scikit-learn, LightGBM, CatBoost, Ensemble Methods, Feature Engineering, Conformal Prediction

Development & Deployment: Docker, FastAPI, React, Streamlit, REST APIs, Git, CI/CD, GitHub Actions, pytest, Nginx

Spoken Languages: English (Native), Arabic (Bilingual), French (Elementary) | U.S. Citizen